

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

Annexure A

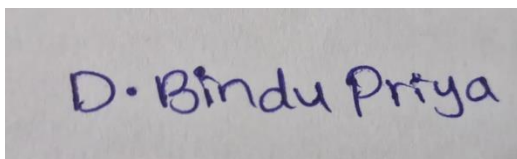
Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I D.BINDU PRIYA(192425162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

1. A Neural Network Predicts Student Scores with an MSE of 0.85 During Training and 5.20 During Testing

Error Identification

The neural network achieves a **Mean Squared Error (MSE) of 0.85** on the training dataset but **5.20** on the testing dataset. The large difference between the training and testing errors indicates that the model performs very well on the training data but poorly on unseen test data. This is a clear indication of **overfitting**, where the model memorizes the training data instead of learning general patterns.

The identified errors are:

- High testing error compared to training error.
- Poor generalization of the model.
- Possible overfitting of the neural network.

Error Analysis & Reasoning

The significant increase in testing MSE may occur due to the following reasons:

1. Overfitting

The neural network has learned the training data too closely, including noise and unnecessary details. As a result, it cannot accurately predict new data.

2. Complex Model Architecture

The model may contain too many hidden layers or neurons, making it excessively complex for the available dataset.

3. Insufficient Training Data

If the dataset is small, the neural network may fail to learn the overall data distribution, resulting in poor testing performance.

4. Lack of Regularization

Without regularization techniques such as **L1**, **L2**, or **Dropout**, the model tends to overfit the training data.

5. Data Distribution Difference

The training and testing datasets may not have similar characteristics. Different distributions reduce prediction accuracy on unseen data.

6. Poor Data Preprocessing

Improper normalization, missing values, or noisy data can negatively affect the testing performance.

Thus, the large difference between the two MSE values clearly suggests that the model is not generalizing well to unseen student data.

Correction Strategy

The following corrective measures can improve the model performance:

1. Apply Regularization

Use **L1 or L2 regularization** to reduce model complexity and prevent overfitting.

2. Introduce Dropout Layers

Dropout randomly disables neurons during training, forcing the network to learn more generalized features.

3. Increase Training Data

Collect additional student performance data or apply data augmentation techniques to improve learning.

4. Simplify the Neural Network

Reduce the number of hidden layers or neurons to avoid memorizing the training data.

5. Use Early Stopping

Monitor the validation loss and stop training once it begins to increase, preventing unnecessary learning of noise.

6. Perform Proper Data Preprocessing

Normalize input features, remove outliers, handle missing values, and ensure that training and testing datasets follow similar distributions.

7. Apply Cross-Validation

Use **k-fold cross-validation** to evaluate the model on different subsets of data and improve its reliability.

Interpretation of Results

The **training MSE (0.85)** indicates that the model fits the training data very well. However, the **testing MSE (5.20)** is much higher, demonstrating poor

prediction accuracy on unseen student data. This performance gap confirms that the model suffers from **overfitting** rather than underfitting.

By applying techniques such as **regularization, dropout, early stopping, simplifying the model architecture, increasing the dataset size, and proper preprocessing**, the testing error can be reduced while maintaining good training performance. As a result, the neural network will achieve better **generalization**, higher prediction accuracy, and more reliable performance when predicting student scores in real-world scenarios.